17/08/20

1. $4te^t$

$$\boxed{L\{t\,f(t)\} = -F'(s)}$$

$$F(s) = \frac{1}{s-1}$$

$$F'(s) = \frac{-1}{(s-1)^2}$$

$$F\{4t\,f(t)\} = \frac{+4}{(s-1)^2}$$

2. $t \sin wt$

$$L\{f(t)\} : F(s) = \frac{w}{s^2+w^2}$$

$$F'(s) = \frac{(s^2+w^2)\,0 - w(2s)}{(s^2+w^2)}$$

$$= \frac{2ws}{(s^2+w^2)^2}$$

$$L\{t\,f(t)\} = \frac{+2ws}{(s^2+w^2)^2}.$$

3. $-t \cosh 2t$.

$$f(t) = -\cosh 2t$$

$$= \frac{-s}{s^2-4}$$

$$= -\frac{[s^2-4 - s\,2s]}{(s^2-4)^2}$$

$$L\{t\,f(t)\} = -\left[\frac{-[s^2-4-2s^2]}{(s^2-4)^2}\right] = \frac{-s^2-4}{(s^2-4)^2} = \frac{-1}{(s^2-4)}$$

5. $f(t) = e^{-2t} \sin t$.

Shifting theorem

$$\frac{1}{s^2+1} \quad \left(\frac{1}{(s+2)^2+1}\right)' = -\frac{2(s+2)}{((s+2)^2+1)^2}$$

$$L\{t f(t)\} = -F'(s)$$

$$= \frac{2(s+2)}{((s+2)^2+1)^2}$$

6. $t^2 \sinh 4t$.

$$L\{t^2 (f(t))\} = \frac{d^2}{ds^2} F(s).$$

$$f(t) = \sinh 4t$$

$$L\{f(t)\} = \frac{4}{s^2-16}$$

$$F'(s) = \frac{-1}{(s^2-16)^2}(2s)\ 4$$

$$F''(s) = \frac{(s^2-16)^2(-2) + 2s(2(s^2-16)2s)}{(s^2-16)^4}$$

$$8\left[\frac{3s^2+16}{(s^2-16)^3}\right] \quad = \frac{-2s^2}{}$$

7 $t^2 \sin \omega t$.

$$F(s) = \frac{\omega}{s^2+\omega^2}$$

$$F'(s) = \frac{-\omega}{(s^2+\omega^2)^2} 2s \quad \frac{-\omega 2s}{(s^2+\omega^2)^2}$$

$$-\omega \frac{\left[(s^2+\omega^2)^2 2 - 2s(2(s^2+\omega^2)2s)\right]}{(s^2+\omega^2)^4}$$

$$\frac{-2\omega\left[(s^2+\omega^2) - (4s^2)\right]}{(s^2+\omega^2)^3} \quad \left[\frac{6\omega s^2 - 2\omega^3}{(s^2+\omega^2)^3}\right]$$

$$\frac{(\omega^2-3s^2)\cdot 2\omega^3}{(s^2+\omega^2)^3}$$

9. $t \cos \omega t$

$$F(s) = \frac{s}{s^2+\omega^2}$$

$$F'(s) = \frac{s^2+\omega^2 - s\,2s}{(s^2+\omega^2)^2} = \frac{\omega^2 - s^2}{(\omega^2+s^2)^2}$$

$$s^2 - \omega^2 / (\omega^2+s^2)^2$$

10. $t\, e^{-kt} \sin t$

$$\frac{\omega}{(s+k)^2+\omega^2}$$

$$F'(s) = \frac{-\omega\; 2(s+k)}{((s+k)^2+\omega^2)^2} = \frac{2(k+s)}{((s+k)^2+1)^2}$$

1. $\ln\left(1 - \frac{a^2}{s^2}\right)$

$$\ln s^2 - a^2 - \ln s^2$$

$$F(s) = \frac{1}{s^2-a^2}\,2s - \frac{1}{s^2}\,2s$$

$$f(t) = \frac{-1}{t}(2\cosh at - 2)$$

2. $\ln \frac{s}{s-1}$

$$F(s) = \frac{1}{s} - \frac{1}{s-1}$$

$$f(t) = (1 - e^{t})\,{}^{-1}/_{t}$$

3. $\ln\left(\frac{s+a}{s+b}\right)$

$$F'(s) = \frac{1}{s+a} - \frac{1}{s+b}$$

$$= \frac{-1}{t}\left(e^{-at} - e^{-bt}\right)$$

4. $\gamma(\cot s/\omega)$ =

$$F'(s) = -\frac{1}{\sin^2 s/\omega} \times 1/\omega$$

$$= \left(+\frac{1}{t\omega \sin^2 s/\omega}\right)$$

$$\frac{1}{t\omega} \operatorname{cosec}^2 s/\omega$$

$$\cot^{-1} s/\omega = \frac{-1}{1+\frac{s^2}{\omega^2}} = \frac{-\omega^2}{\omega^2+s^2}$$

$$\frac{\omega^2}{(\omega^2+s^2)} \quad (\omega \sin \omega t) \; +1/t$$

4. $L\{t\cos(t+k)\} = L\{t\cos t\cos k - \sin t \sin k\}$

$$= \frac{-d}{ds}\left[\frac{\cos k \cdot s}{s^2+1} - \frac{\sin k}{s^2+1}\right]$$

$$= \left[\frac{\cos k\,(1-s^2) + \sin k\, 2s}{(s^2+1)^2}\right]$$

$$= \frac{\cos k\,(s^2-1) - 2s\sin k}{(s^2+1)^2}$$

8. $L\{t \sin(t+1)t\}$

$$= \frac{-d}{ds}\left[\frac{\cos k}{s^2+1} + \frac{s\sin k}{s^2+1}\right]$$

$$= \frac{2s\cos k + \sin k\,(1-s^2)}{(s^2+1)^2}$$

$$\frac{2s\cos k + (s^2-1)\sin k}{(s^2+1)^2}$$

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